

module3_lab3

September 17, 2026

```
[1]: import pandas as pd
import numpy as np

df = pd.read_csv('bridges.csv')
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")
df.head()
```

Rows: 108, Columns: 13

```
[1]:   Id river  location  erected  purpose  length  lanes  clear-g  t-or-d \
0  E1     M      3.0    1818   HIGHWAY    NaN     2.0      N   THROUGH
1  E2     A     25.0    1819   HIGHWAY  1037.0     2.0      N   THROUGH
2  E3     A     39.0    1829  AQUEDUCT    NaN     1.0      N   THROUGH
3  E5     A     29.0    1837   HIGHWAY  1000.0     2.0      N   THROUGH
4  E6     M     23.0    1838   HIGHWAY    NaN     2.0      N   THROUGH

   material  span rel-l  type
0     WOOD  SHORT     S  WOOD
1     WOOD  SHORT     S  WOOD
2     WOOD   NaN     S  WOOD
3     WOOD  SHORT     S  WOOD
4     WOOD   NaN     S  WOOD
```

0.1 1. Look at the data before assuming anything

length is the only numeric attribute in this dataset (feet). Everything else (river, purpose, material, span, type, etc.) is categorical or an identifier. Before stating a plausibility rule, it's worth knowing (a) how much data is missing and (b) the overall shape of the distribution — a rule should be informed by what “normal” looks like for *these* bridges, not an arbitrary number.

```
[2]: missing = df['length'].isna().sum()
print(f"Missing 'length' values: {missing} of {len(df)} rows ({missing/len(df):.
    ↪1%})")
print()
df['length'].describe()
```

Missing 'length' values: 27 of 108 rows (25.0%)

```
[2]: count      81.000000
     mean     1567.469136
     std       747.491523
     min       804.000000
     25%      1000.000000
     50%      1300.000000
     75%      2000.000000
     max      4558.000000
     Name: length, dtype: float64
```

Two things worth noting:

- **27 rows (25%) have no recorded length at all.** These are *missing* values, not outliers — there is nothing implausible about a blank cell, so they are not part of the outlier screen below. They are a separate data-quality issue (handled at the end).
- Among the 81 rows that do have a length, the values range from 804 ft to 4,558 ft, with a median around 1,300 ft. No length is zero or negative, so there’s no “obviously impossible” value the way Lab 2 had — this is exactly the subtler situation the assignment describes.

0.2 2. Stated assumption (before screening)

Context: these are 19th- and early-20th-century highway and railroad bridges in the Pittsburgh area, crossing the Allegheny, Monongahela, and Ohio rivers. `length` is the total structure length in feet.

Plausibility rule, stated before I look at what it flags:

A bridge `length` is plausible if it falls between **200 ft and 3,000 ft**.

- **Lower bound (200 ft):** a structure recorded in this dataset as a “bridge” spanning a river should be at least long enough to actually cross open water — a few dozen feet would suggest a culvert or a recording error, not a river bridge.
- **Upper bound (3,000 ft):** the vast majority of 19th-century wood/iron/steel truss and suspension bridges in this region were in the few-hundred-to-low-thousands-of-feet range. 3,000 ft is a generous ceiling — well above the 75th percentile (2,000 ft) — chosen so that only *unusually* long structures get flagged, not just the long tail of an already-long-tailed distribution.

Rows outside [200, 3000] are flagged below. I am committing to this rule **before** running the screen, per the assignment’s instructions.

```
[3]: LOWER_BOUND = 200
     UPPER_BOUND = 3000

     has_length = df['length'].notna()
     implausible = has_length & ((df['length'] < LOWER_BOUND) | (df['length'] >
     ↪UPPER_BOUND))

     print(f"Rows screened (non-missing length): {has_length.sum()}")
     print(f"Rows flagged as implausible: {implausible.sum()}")
```

```
print()
flagged = df.loc[implausible, ['Id', 'river', 'erected', 'purpose', 'length',
↪ 'material', 'span', 'type']] \
    .sort_values('length', ascending=False)
flagged
```

Rows screened (non-missing length): 81

Rows flagged as implausible: 3

```
[3]:      Id river  erected  purpose  length material  span      type
31   E34      0    1888      RR  4558.0    STEEL  LONG  SIMPLE-T
44   E46      A    1897      RR  4000.0    STEEL  LONG  SIMPLE-T
104  E91      0    1975  HIGHWAY  3756.0    STEEL  LONG    ARCH
```

0.3 3. What got flagged

Three rows exceed the 3,000 ft upper bound: **E34** (4,558 ft), **E46** (4,000 ft), and **E91** (3,756 ft). No rows fall below the 200 ft lower bound — every recorded length is at least 804 ft, so the lower bound did not end up screening anything out for this particular dataset (it’s still correct to state it, since a rule chosen *after* seeing that no row is short would be reasoning backwards).

```
[4]: # A closer look at why the three flagged rows are so long
flagged[['Id', 'river', 'erected', 'purpose', 'material', 'span', 'type',
↪ 'length']]
```

```
[4]:      Id river  erected  purpose material  span      type  length
31   E34      0    1888      RR    STEEL  LONG  SIMPLE-T  4558.0
44   E46      A    1897      RR    STEEL  LONG  SIMPLE-T  4000.0
104  E91      0    1975  HIGHWAY    STEEL  LONG    ARCH   3756.0
```

0.4 4. Handling the flagged rows, and why

I am **not deleting or capping** these three rows. Here’s the reasoning:

- All three (E34, E46, E91) have `span == LONG` and `material == STEEL`. That combination is exactly what you’d expect to *legitimately* produce a length far outside the typical wood/iron truss range: a long-span steel bridge (e.g. a railroad bridge crossing the Ohio River, or the 1975 highway bridge E91) is structurally capable of — and expected to — run several thousand feet, especially for a multi-span railroad crossing.
- In other words, my *statistical* rule (3,000 ft ceiling) correctly flagged these as unusual relative to the rest of the dataset, but the *domain* evidence in the other columns (`span`, `material`, `purpose`) supports treating them as real, valid measurements rather than data-entry errors — there’s no column where a “4,558” could plausibly be a typo for something in-range (e.g. a decimal-point slip would need to be off by 10x, which isn’t a typical typo pattern).
- **Decision:** keep all three rows unchanged in the analysis dataset, but retain the boolean `length_flag` column below so any downstream analysis (e.g. an average-length-by-material calculation) can choose to exclude or separately weight them if their scale would distort a mean or a chart.

Separately, the **27 missing length values** are handled differently, since they are a missing-data problem rather than an outlier problem: I leave them as NaN (not imputed to 0 or to the mean) so that any later aggregation on `length` correctly excludes them via `.dropna()/.mean()` rather than being silently biased by a fabricated fill value.

```
[5]: df['length_flag'] = implausible
print("Value counts for the new 'length_flag' column:")
print(df['length_flag'].value_counts())
print()
kept = (~df['length_flag']).sum()
print(f"Rows kept as-is (not flagged, including the 27 with missing length): ␣
↪{kept} / {len(df)}")
```

Value counts for the new 'length_flag' column:

```
length_flag
False      105
True         3
Name: count, dtype: int64
```

Rows kept as-is (not flagged, including the 27 with missing length): 105 / 108

0.5 5. Summary

- **Assumption (stated first):** a plausible bridge length is between 200 ft and 3,000 ft, based on the physical need to span a river and the typical scale of 19th-century truss/suspension bridges in this region.
- **Screen:** applied to the 81 non-missing `length` values; flagged 3 of them (E34, E46, E91), all > 3,000 ft.
- **Handling:** kept all three flagged rows — the domain columns (`span=LONG`, `material=STEEL`) support them as genuine long-span steel bridges rather than data-entry errors — and added a `length_flag` column so they can be excluded from length-sensitive summary statistics later if needed. The 27 missing lengths are left as NaN, since they are a separate missing-data issue, not an outlier.

GitHub repo: <https://github.com/roee0910/cs82a-portfolio.git>